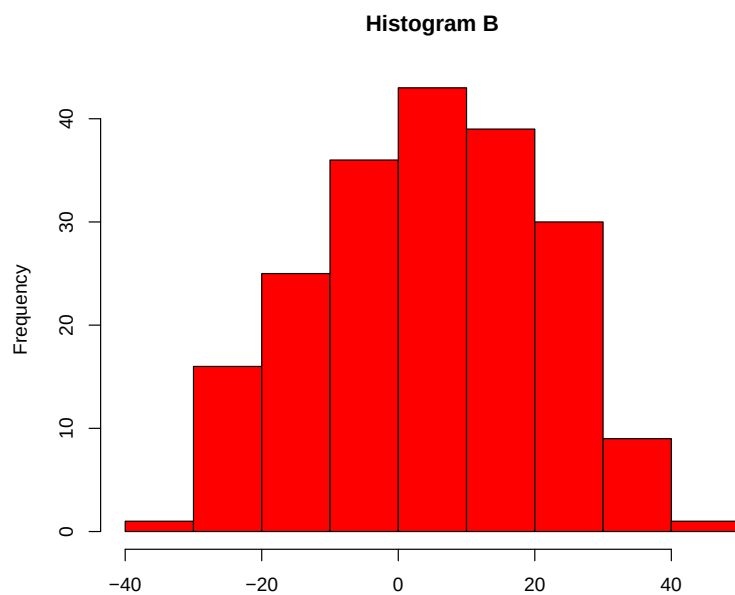
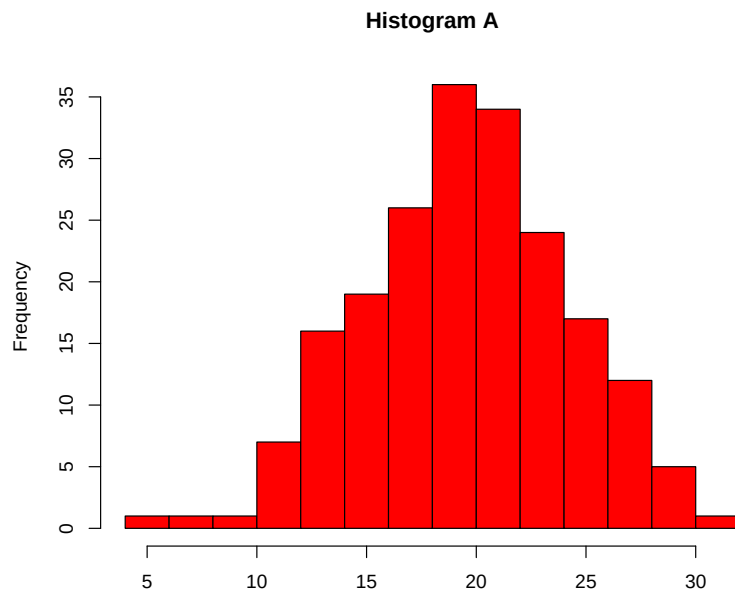


Homework 2

Write a `HW2.Rmd` file to answer these questions. Knit your `HW2.Rmd` to `HW2.html` along the way. Turn in `HW2.html` to Canvas.

1. Consider the data in the `spider.csv` file on the running speed (cm/s) of a certain species of spider.
 - (a) Use R to create a frequency histogram of the data.
 - (b) Use a calculator to compute the sample mean and sample standard deviation for the data. Then use R to compute them. (Check that your calculator and R give the same answers.) Interpret these values—that is, what does the mean indicate about the data and what does the standard deviation indicate?
 - (c) Use the rules in the lecture notes to find the three quartiles for the data. Then use R to compute them. Do the answers agree?
 - (d) Use R to compute the range and IQR for the data.
 - (e) Make a boxplot of the data by hand. (You do not need to show evidence of this in your `.Rmd` file.) Make the boxplot with R to check your work; include the R boxplot here.

2. Consider the following two histograms:



- (a) Say which of the two histograms has a larger mean, or if they are about the same. Explain your answer.
- (b) Say which of the two histograms has a larger standard deviation, or if they are about the same. Explain your answer.
- (c) What simple change to this pair of graphs would have made the previous two questions easy to answer?